MUHAMMED SHEHEEN M T P

AI ENGINEER | AI AUTOMATION | PYTHON BACKEND DEVELOPMENT

+971 521823134 | Abu Dhabi, UAE

muhammedsheheen0@gmail.com | linkedin.com/in/sheheen-mtp/ | github.com/sheheenmtp | sheheenmtp.vercel.app

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in Python application development, backend engineering, LLM integration, NLP, and workflow automation. Experienced in developing RESTful APIs and backend services using FastAPI, PostgreSQL, SQLAlchemy, and Docker, with practical implementation of adaptive prompting, local LLM integration using Ollama, and sandboxed code execution using Judge0. Contributed to Malayalam NLP data engineering, English–Malayalam parallel corpus preparation, machine-translation workflows, and ASR/OCR data quality assurance at ICFOSS. Also experienced in translating business and operational requirements into practical automation solutions using Google Apps Script, Google Sheets, Forms, and Drive for booking management, client records, and payment tracking. Currently developing an AI-powered job-search automation system using n8n and PostgreSQL, focusing on modular workflows for job discovery, data extraction, verification, and candidate–job matching. Focused on building practical AI applications, reliable backend services, and workflow automation solutions.

TECHNICAL SKILLS

Programming & Backend: Python, SQL, JavaScript, FastAPI, REST APIs, SQLAlchemy

AI & LLM: LLM integration, prompt engineering, adaptive prompting, Ollama, qwen2.5-coder:14b

ML & NLP: PyTorch, Hugging Face Transformers, NLLB-200, PEFT/LoRA, machine translation, NLP, BLEU, chrF

Data & Execution: PostgreSQL, Docker, ONNX Runtime, Judge0

Automation & Tools: n8n, Google Apps Script, Google Workspace, Git, GitHub, Linux, Jupyter, Google Colab

Computer Vision: OpenCV, YOLOv8, MediaPipe

PROFESSIONAL EXPERIENCE

ICFOSS — AI Engineer **Aug 2025 – Aug 2026 | Thiruvananthapuram, Kerala, India**

- Developed and deployed AI Tutor backend services with Python, FastAPI, PostgreSQL, SQLAlchemy, and Docker; implemented 15 REST endpoints for authentication, learner progress, recommendations, code evaluation, and AI feedback.
- Integrated Ollama (qwen2.5-coder:14b) for adaptive coding assistance and Judge0 for sandboxed multi-language execution; implemented learner-aware recommendations using learner weaknesses, retry patterns, learning progression, spaced practice, anti-repetition logic, and repeated-failure cooldowns
- Contributed to BHASHINI-related Malayalam NLP data engineering, English–Malayalam parallel-data preparation, OCR training-data preparation, ASR transcription QA, and dialect documentation.
- Supported live speech recognition and multilingual translation at the VISION 2031 International Conference, including audio integration, troubleshooting, and latency monitoring.

XOGRID — Technical Consultant **Aug 2025 – Oct 2026**

- Translated business and operational requirements into a Google Workspace booking and client-management workflow using Google Apps Script, JavaScript, Sheets, Forms, and Drive.
- Automated client intake, booking and availability records, seat/cabin allocation, client documents, monthly processing, renewal matching, and payment-status tracking.
- Configured scheduled and Google Sheets edit-triggered scripts; investigated payment-cycle and date-boundary edge cases to improve record consistency.

SELECTED TECHNICAL PROJECTS

AI Tutor Platform | Python, FastAPI, PostgreSQL, SQLAlchemy, Ollama, Judge0, React/Vite, Docker

- Developed a full-stack AI-powered coding platform prototype with authentication, REST APIs, learner progress tracking, problem recommendations, code evaluation, and AI-generated feedback.
- Integrated Ollama (qwen2.5-coder:14b) for adaptive coding assistance and Judge0 for sandboxed code execution, with prompts tailored to learner proficiency.
- Designed a personalized problem-recommendation algorithm using concept weaknesses, retry patterns, learning progression, spaced practice, and anti-repetition logic.

English–Malayalam Neural Machine Translation | Python, PyTorch, Hugging Face Transformers, NLLB-200 Distilled 600M, PEFT/LoRA, FastAPI, ONNX Runtime

- Prepared and cleaned English–Malayalam parallel datasets, fine-tuned NLLB-200 Distilled 600M using LoRA, and evaluated translation quality using BLEU and chrF metrics.
- Merged the fine-tuned adapter with the base model, exported the model to ONNX Runtime, and developed a local FastAPI service for machine-translation inference.

XO GRID Booking Management Automation | Google Apps Script, JavaScript, Google Workspace

- Developed automated booking and client-management workflows using Google Apps Script, Google Sheets, Forms, and Drive, streamlining client registration, workspace allocation, and booking administration.
- Implemented scheduled and edit-triggered automation for monthly record processing, renewal matching, payment-status tracking, and record synchronization, with diagnostic functions to support troubleshooting and data consistency.

AI-Powered Job Search Automation — ACTIVE DEVELOPMENT | self-hosted n8n, PostgreSQL

- Developing an AI-powered job search automation system using self-hosted n8n and PostgreSQL, with modular workflows for job discovery, data extraction, verification, candidate–job matching, and application tracking; the job-discovery workflow is operational, while extraction and SQL-processing stages remain under development.

Autonomous USV for Aquatic Waste Detection | Python, OpenCV, YOLOv8, ESP32-WROOM-32, WebSocket, IP Webcam

- Developed a university prototype using live video and YOLOv8-based object detection to identify floating plastic waste in aquatic environments.
- Integrated computer vision with ESP32-based motor control and WebSocket communication to support autonomous navigation and remote operation.

EDUCATION

Bachelor of Technology in Information Technology **2021 - 2025**
— College of Engineering Thalassery, Kerala, India

Diploma in Mechanical Engineering **2018-2021**
— Swami Nithyananda Polytechnic College,
Kanhangad, Kerala, India

CERTIFICATIONS

Generative AI: Prompt Engineering Basics — IBM / Coursera
Generative AI: Introduction and Applications — IBM / Coursera
Introduction to Artificial Intelligence — IBM / Coursera
Python for Data Science, AI and Development — IBM / Coursera
IBM AI Foundations for Business Specialization
PostgreSQL for Everybody Specialization — University of Michigan / Coursera